

ECO101

Introduction to Microeconomics

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Summer 2026



Credit: Cengage 2014

Week 2 - Trade

- *Why do people—and nations—choose to be economically interdependent?*
- *How can trade make everyone better off?*
- *What is absolute advantage? What is comparative advantage? How are these concepts similar? How are they different?*

Interdependence

One of the Principles from the first lecture:

Trade can make everyone better off.

We now learn why people—and nations—choose to be interdependent, and how they can gain from trade.

Interdependence

Two countries: the U.S. us and Japan JP

Two goods: computers  and wheat 

One resource: labor  , measured in hours

We will look at how much of both goods each country produces and consumes

- if the country chooses to be self-sufficient
- if it trades with the other country

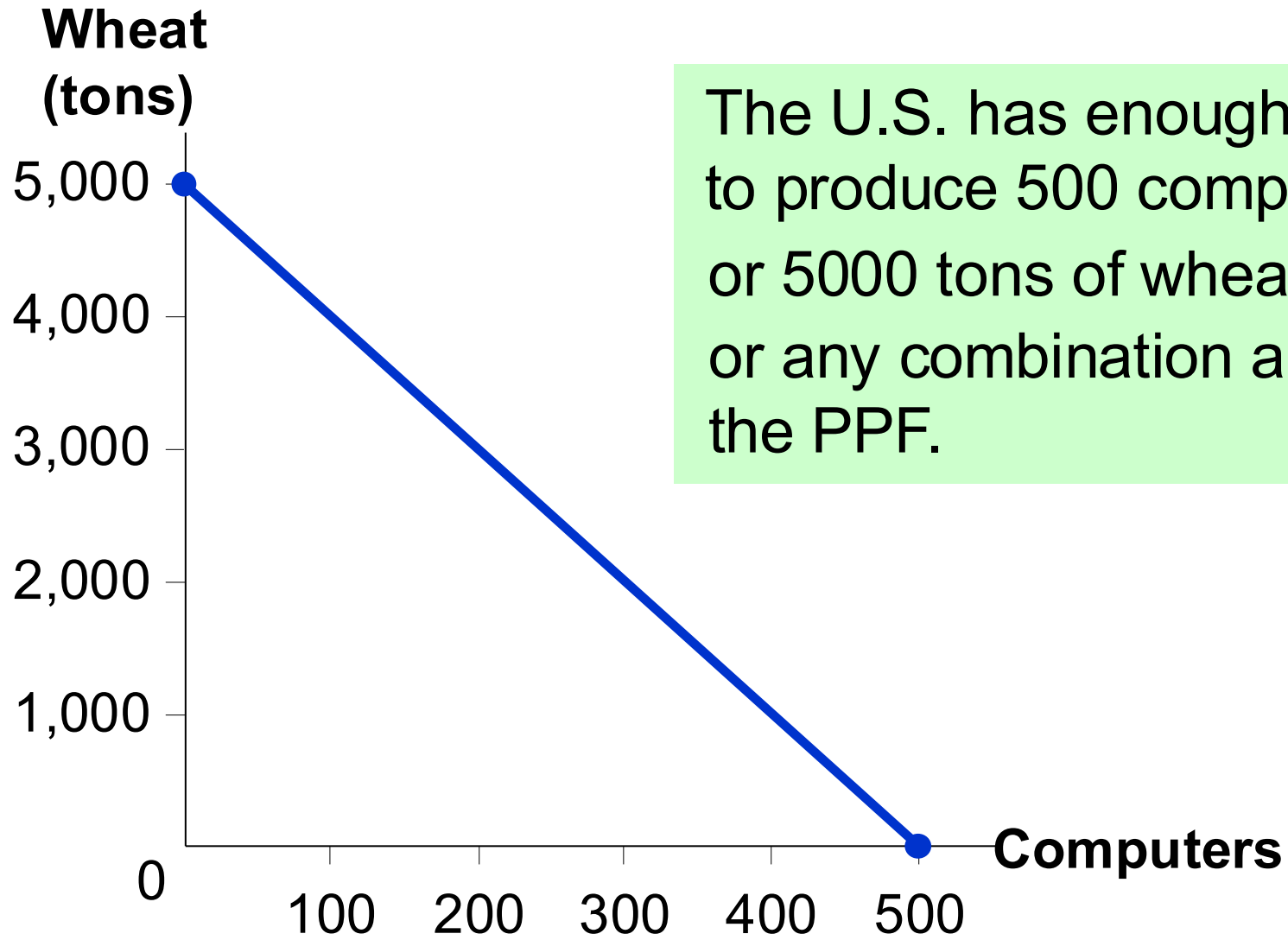
Production Possibilities in the U.S.

The U.S. has 50,000 hours of labor available for production, per month.

Producing one computer requires 100 hours of labor.

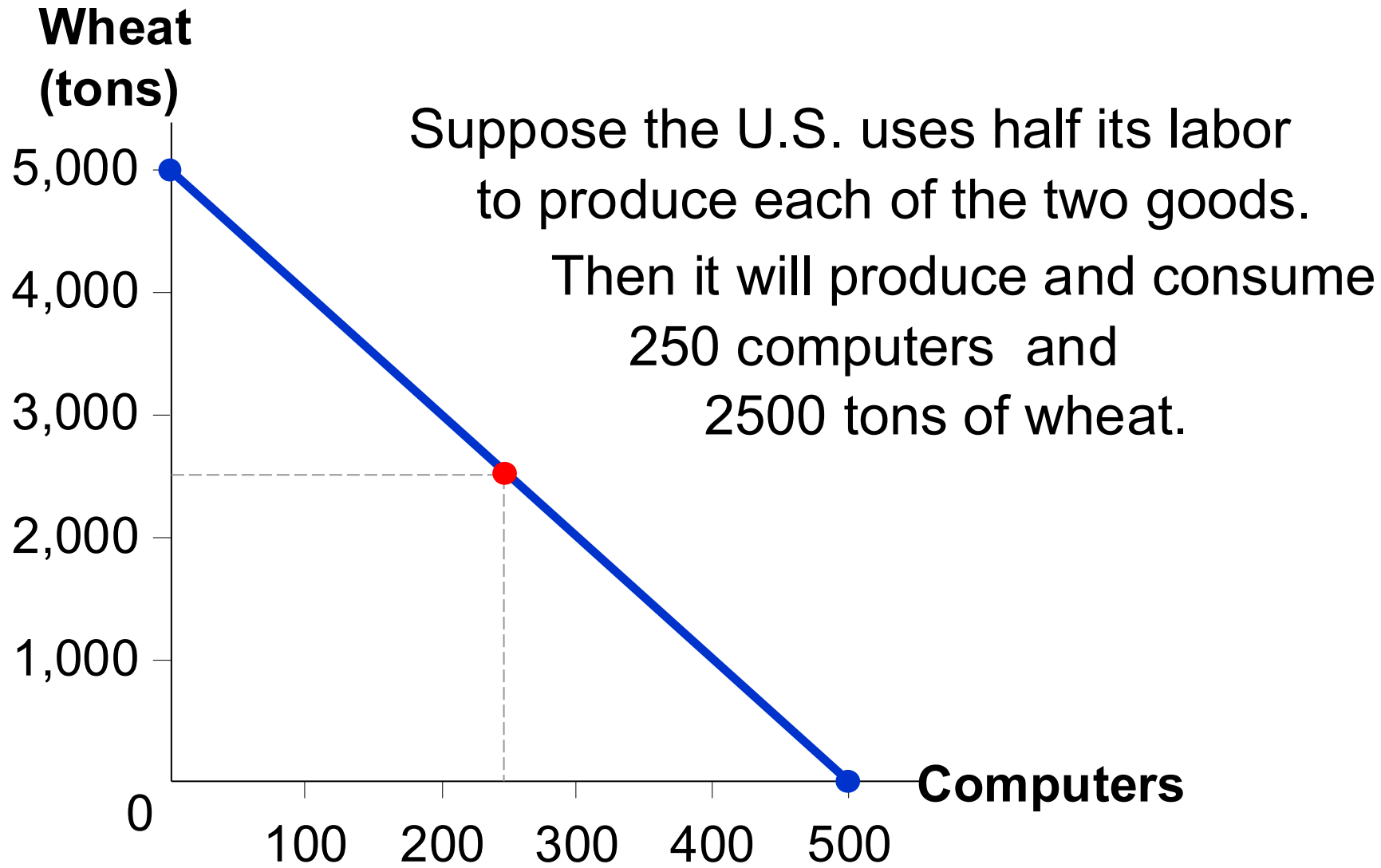
Producing one ton of wheat requires 10 hours of labor.

The U.S. PPF



The U.S. has enough labor to produce 500 computers, or 5000 tons of wheat, or any combination along the PPF.

The U.S. Without Trade



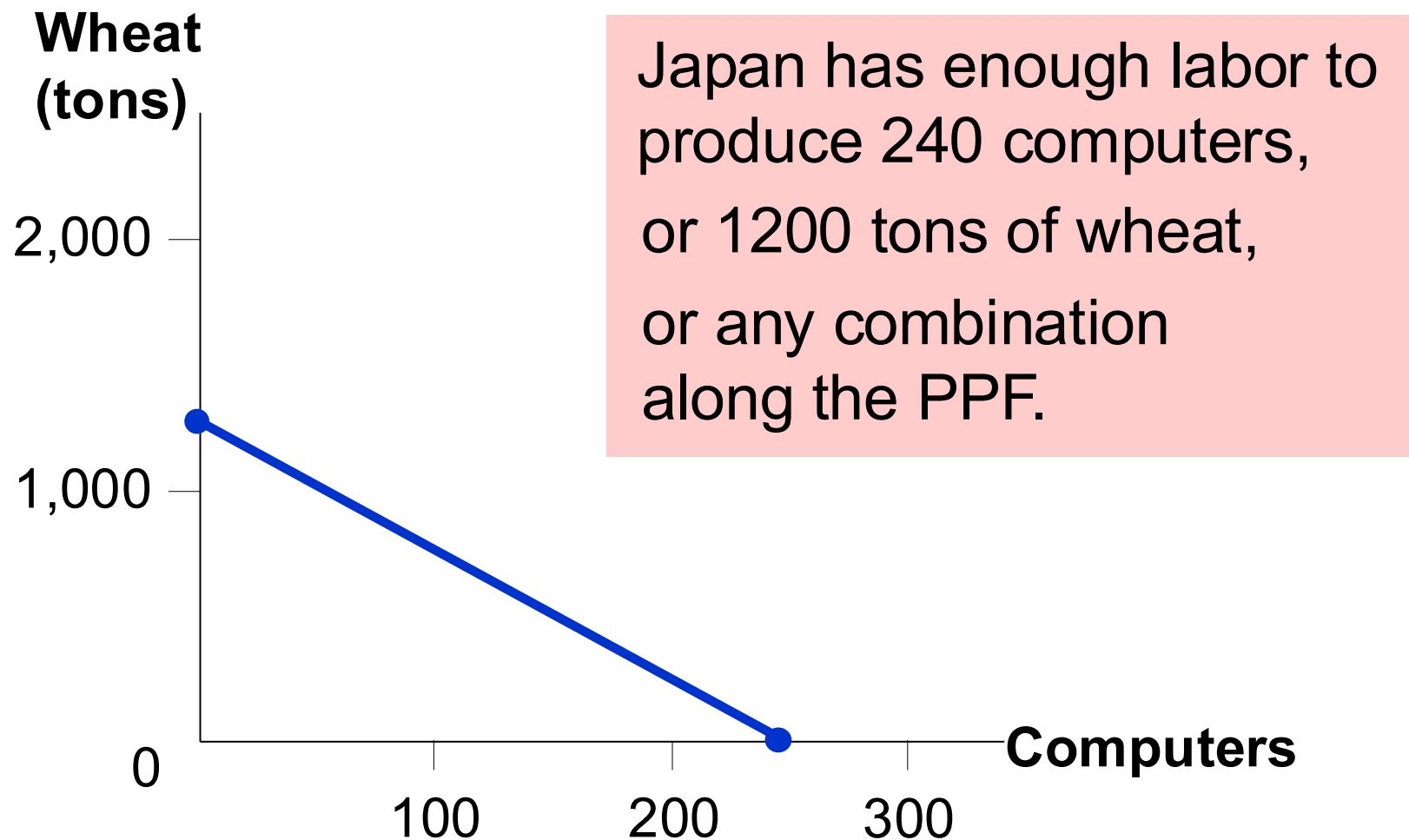
Production Possibilities in Japan

Japan has 30,000 hours of labor available for production, per month.

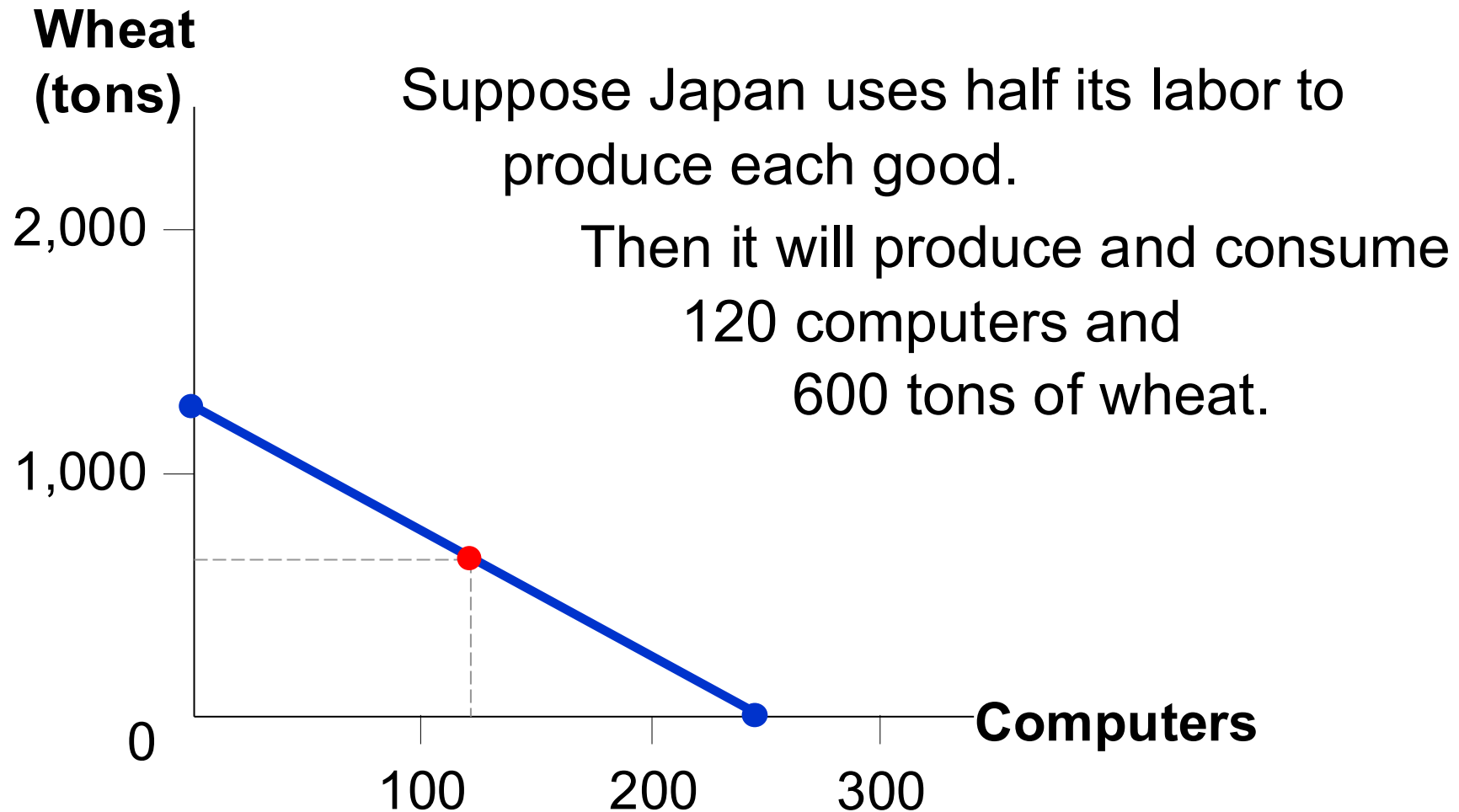
Producing one computer requires 125 hours of labor.

Producing one ton of wheat requires 25 hours of labor.

Japan's PPF



Japan Without Trade



Consumption With and Without Trade

Without trade,

us consumers get 250 🖥️ and 2500 tons 🌽.

JP consumers get 120 🖥️ and 600 tons 🌽.

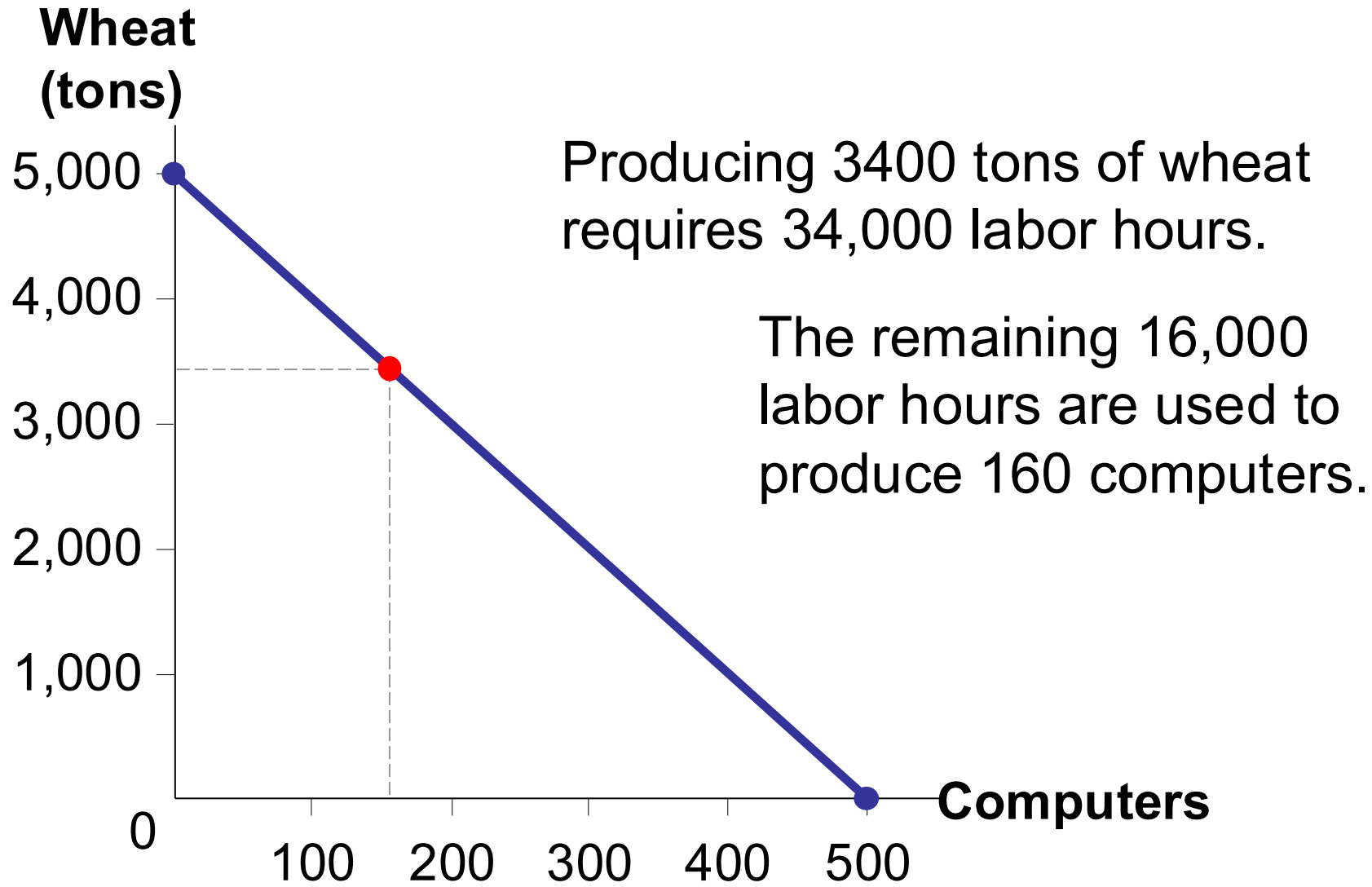
We will compare consumption without trade to consumption with trade.

First, we need to see how much of each good is produced and traded by the two countries.

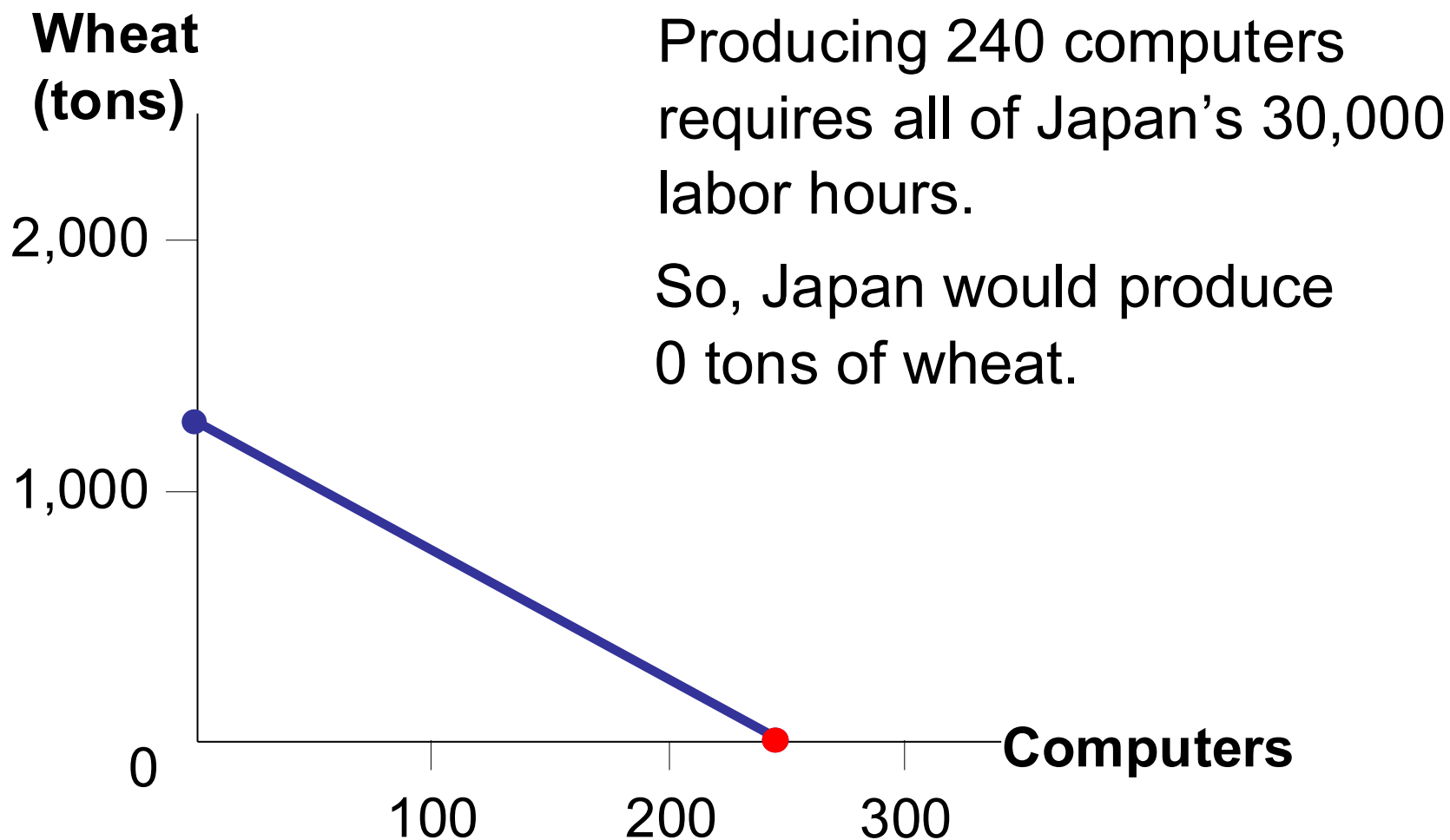
Consumption With and Without Trade

1. Suppose the us produces **3400 tons of** 🌽. How many 🖥️ would the us be able to produce with its remaining labor?
2. Suppose JP produces **240** 🖥️. How many tons of 🌽 would JP be able to produce with its remaining labor?

U.S. Production With Trade



Japan's Production With Trade



Exports & Imports

Exports: goods produced domestically and sold abroad

To export means to sell domestically produced goods abroad.

Imports: goods produced abroad and sold domestically

To import means to purchase goods produced in other countries.

Consumption under Trade

Suppose the U.S. exports 700 tons of 🌽 to Japan, and imports 110 🖥️ from Japan.

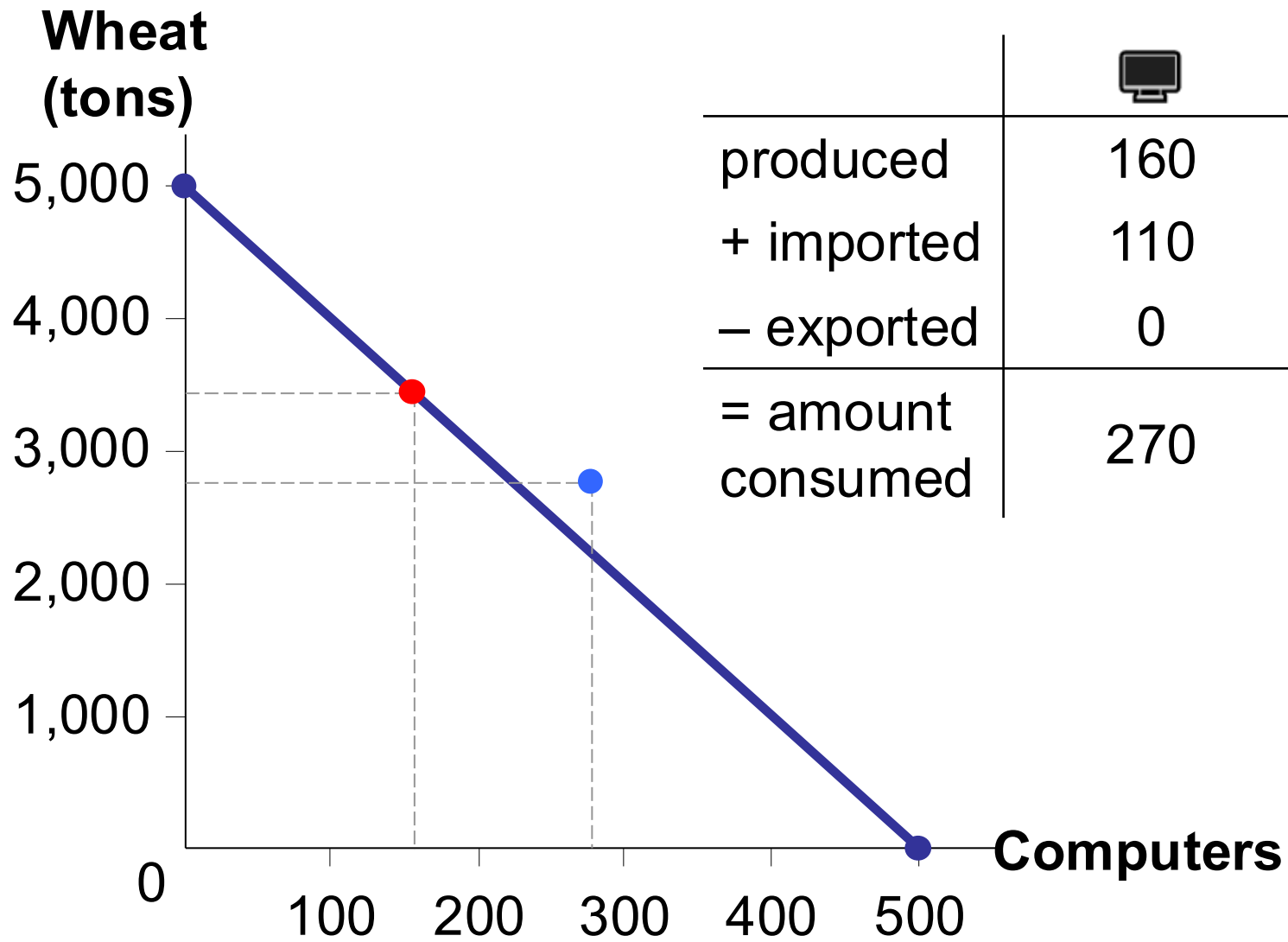
(So, Japan imports 700 tons 🌽 and exports 110 🖥️.)

How much of each good is consumed in the U.S.?

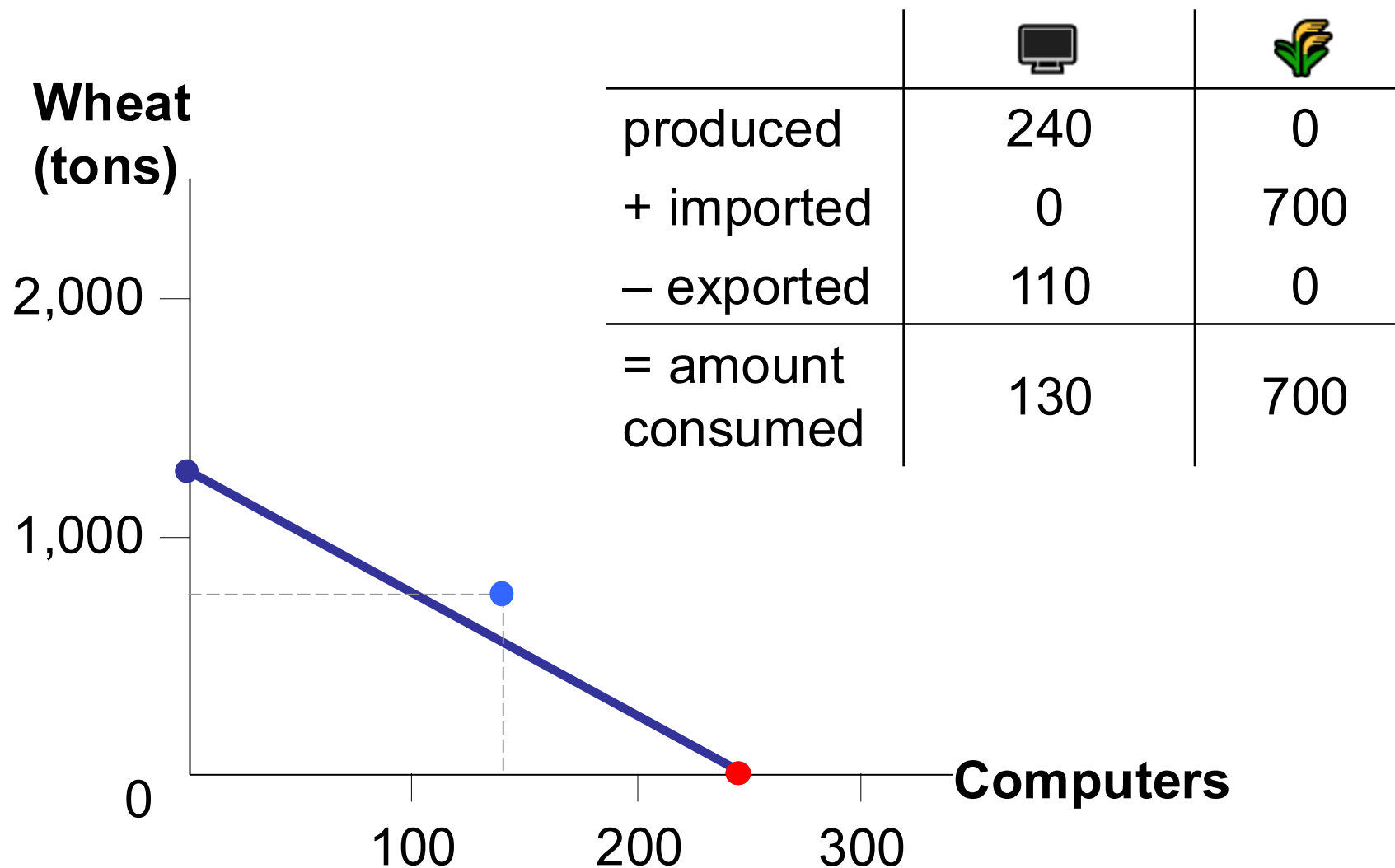
How much of each good is consumed in Japan?

US

U.S. Consumption With Trade



Japan's Consumption With Trade



Trade Makes Both Countries Better Off

US			
	consumption without trade	consumption with trade	gains from trade
	250	270	20
	2500	2700	200
JP			
	consumption without trade	consumption with trade	gains from trade
	120	130	10
	600	700	100

Where Do These Gains Come From?

Absolute advantage: the ability to produce a good using fewer inputs than another producer

The U.S. has an absolute advantage in **wheat**: *producing a ton of wheat uses 10 labor hours in the U.S. vs. 25 in Japan.*

If each country has an absolute advantage in one good and specializes in that good, then both countries can gain from trade.

Where Do These Gains Come From?

Which country has an absolute advantage in computers?

Producing one computer requires 125 labor hours in Japan, but only 100 in the U.S.

The U.S. has an absolute advantage in both goods!

So why does Japan specialize in computers?

Why do both countries gain from trade?

Two Measures of the Cost of a Good

Two countries can gain from trade when each specializes in the good it produces at lowest cost.

Absolute advantage measures the cost of a good in terms of the inputs required to produce it.

Recall:

Another measure of cost is *opportunity cost*.

In our example, the opportunity cost of a computer is the amount of wheat that could be produced using the labor needed to produce one computer.

Opportunity Cost and Comparative Advantage

Comparative advantage: the ability to produce a good at a lower opportunity cost than another producer

Which country has the comparative advantage in computers?

To answer this, we must determine the opportunity cost of a computer in each country.

Opportunity Cost and Comparative Advantage

The opportunity cost of a computer is

10 tons of wheat in the U.S., because producing one computer requires 100 labor hours, which instead could produce 10 tons of wheat.

5 tons of wheat in Japan, because producing one computer requires 125 labor hours, which instead could produce 5 tons of wheat.

So, Japan has a comparative advantage in computers. ***Lesson: Absolute advantage is not necessary for comparative advantage!***

Comparative Advantage and Trade

Gains from trade arise from comparative advantage (differences in opportunity costs).

When each country specializes in the good(s) in which it has a comparative advantage, total production in all countries is higher, the world's "economic pie" is bigger, and all countries can gain from trade.

Unanswered Questions...

We made a lot of assumptions about the quantities of each good that each country produces, trades, and consumes, and the price at which the countries trade wheat for computers.

In the real world, these quantities and prices would be determined by the preferences of consumers and the technology and resources in both countries.

For now, though, our goal was merely to see how *trade can make everyone better off*.

Food for thought

Argentina and Brazil each have 10,000 hours of labor per month.

In Argentina,

producing one pound coffee requires 2 hours

producing one bottle wine requires 4 hours

In Brazil,

producing one pound coffee requires 1 hour

producing one bottle wine requires 5 hours

Which country has an absolute advantage in the production of coffee?

Which country has a comparative advantage in the production of wine?

Food for thought

Brazil has an absolute advantage in coffee:

Producing a pound of coffee requires only one labor-hour in Brazil, but two in Argentina.

Argentina has a comparative advantage in wine:

Argentina's opp. cost of wine is two pounds of coffee, because the four labor-hours required to produce a bottle of wine could instead produce two pounds of coffee.

Brazil's opp. cost of wine is five pounds of

SUMMARY

Comparative advantage means being able to produce a good at a **lower opportunity cost**.

Absolute advantage means being able to produce a good with the **fewest inputs**.

Even if a country has both absolute advantage and comparative advantage in producing a good, trading will help them achieve a better outcome than without.